MILITARY SPECIFICATION

PROPELLANT, DEUTERIUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for gaseous deuterium. Deuterium is a stable isotope of ordinary hydrogen.

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications and Standards. The following documents, of issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-83690 (USAF)	-	Cylinders, Sampling
MIL-V-2/29	-	Valve, Cylinder, Gas, Hydrogen, 3/4 Inch. Inlet Connection, Type 04-351-1, 2200

STANDARDS

MIL-STD-101	- Color Codes for Pipelines and for Compressed Gas Cylinders
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-1411	- Inspection & Maintainance of Compressed Gas Cylinders

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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REGULATIONS

AFR-71-4 - Preparation of Hazardous Materials for Military Air Shipment

2.1.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR 170-190 Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

P 5800.2 Hazardous Materials 1980 Emergency Response Guide Book. (Copies of DOT P 5800.2 may be obtained upon application tot he American Trucking Association Inc., 1616 P Street N.W., Washington DC 20036).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 20402)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-29 - Recommended Practices for Indicating Which Figures are to be Considered Significant in Specified Limiting Values

ASTM-F-307-70 Pressurant Gas Sampling for Gas Analysis

(Copies of ASTM publications may be obtained upon application to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Composition. The composition of the propellant shall conform to the limits listed in Table I when tested in accordance with the applicable test methods.

TABLE I - Composition

Composition	Limits	Test Paragraph
Purity, Percent by Vol, Min	99.5	4.5.4
Hydrogen, Percent by Vol, Max	0.5	4.5.1
Water, Percent by Vol, Max	0.040	4.5.2
Total Hydrocarbons as Methane Percent by Vol, Max	0.00005	4.5.3

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3.2 Limiting Values. The following applies to all specified limits of this specification. For purposes of determining conformance with those requirements, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limitation value, in accordance with the rounding-off method of ASTM Recommended Practice E 29.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for their performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplier and services conform to the prescribed requirements (6.2).

4.1.1 Point of Inspection. Unless otherwise specified, quality conformance tests of the propellant shall be conducted at the site of filling. The quality conformance tests shall be completed prior to shipment of filled containers (see 6.2).

4.2 Classification of Tests. The inspection and testing of the propellant, pressurizing agent shall be classified as quality conformance tests.

4.3 Quality Conformance Tests. Quality conformance tests are defined as tests specified herein to assure conformance to the requirements of Table I and subsection 5.3.6 of this specification. Quality conformance tests shall consist of Individaul Tests (4.3.1) and Sampling Tests (4.3.2).

4.3.1 Individual Tests. Each cylinder shall be subjected to the following tests:

4.3.1.1 Filling Pressure 4.6.1

4.3.1.2 Leakage 4.6.2

4.3.2 Sampling Plan. Quality of the propellant shall be verified by analysis of lot samples as specified in Table II and subjected to tests described in 4.5.

TABLE II, SAMPLING FOR TEST

NUMBER OF CONTAINERS IN LOT	NUMBER OF CONTAINERS TO BE SAMPLED
1	1
2-40	2
41-70	3
71-over	4

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4.3.2.1 Lot. When deuterium cylinders are filled, each set of cylinders charged on the same manifold at the same time shall constitute a lot.

4.3.2.2 Sample. Each sample shall be of sufficient size to conduct all the quality conformance tests as specified herein. Unless otherwise specified, the quality conformance tests shall be made of each required sample (6.2). When required, an equivalent sample shall be forwarded to a laboratory designated by the procuring activity for subjection to the quality conformance tests specified herein.

4.3.2.2.1 Samplers. The sampler shall be in accordance with MIL-C-83690 or functionally equivalent thereto.

4.3.2.2.2 Sampling Methods. Unless otherwise specified, propellant shall be sampled in accordance with ASTM F 307-70.

4.3.2.3 Cylinders. The number of cylinders filled with gaseous propellant selected for sampling from each lot shall be in accordance with Table II. The propellant from each cylinder sampled shall constitute a separate sample.

4.3.2.4 Bulk Transports. A sample shall be taken from each portable tank, cargo tank, or tank car filled with gaseous propellant.

4.4 Rejection. When any sample of the propellant tested in accordance with 4.5 fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected. Disposition of rejected product shall be as specified by the procuring activity (6.2)

4.5 Quality Conformance Tests

4.5.1 The Hydrogen Content. The hydrogen content of the sample shall be determined by any method. In cases of dispute, the mass spectrometric method in 4.5.1.1 shall be used as referee.

4.5.1.1 By mass spectrometer. Hydrogen standards in He shall be used to calibrate the mass spectrometer.

4.5.2 The Water Content. The water content of the sample shall be determined by one of the following methods. In cases of dispute the electrolytic method in 4.5.2.1 shall be used as the referee.

4.5.2.1 By an electrolytic method. Connect the sample container to a pressure regulator which is attached to the electrolytic moisture apparatus (hygrometer). Open the sample container valve and adjust the pressure to the apparatus in accordance with the manufacturer's recommended value. Allow sufficient time for the indicated moisture content to become stable and read the value obtained while using the most sensitive scale setting possible for the moisture content of the sample. The electrolytic moisture apparatus should be set on a range no greater than ten times the specified maximum moisture content.

4.5.2.2 By the Frost (Dew) Point Method

4.5.2.3 By a piezoelectric sorption hygrometer that is set on a range no greater than ten times the specified maximum moisture content.

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4.5.2.4 By an aluminum oxide capacitor-equipped analyzer on a range that is no greater than ten times the specified maximum moisture content.

4.5.3 The Total Hydrocarbons. The total hydrocarbon content, as methane, of the the sample shall be determined by one of the following procedures. In cases of dispute, the flame ionization type analyzer shall be used as the referee (4.5.3.1).

4.5.3.1 By the flame ionization type analyzer. The analyzer shall be calibrated at appropriate intervals by the use of deuterium calibration gas standards (6.5). The range used shall not be greater than ten times the specified maximum total hydrocarbon content expressed as methane.

4.5.3.2 By the gas cell infrared analyzer. The analyzer shall be calibrated at appropriate intervals with methane-in-deuterium standards (6.5) at approximately 3.5 microns. The sensitivity of the analyzer shall be at least 0.1 ppm.

4.5.4 Final Calculations. The deuterium content of the sample shall be calculated by the following formula:

$$\% D_2 = 100 - \sum Ci$$

Where: $\sum Ci$ = the sum of the impurities found in the sample (volume/percent)

4.6 Gas Cylinder Tests.

4.6.1 Filling Pressure. Samples shall be tested for proper filling pressure by attaching a calibrated Bourdon tube or equivalent gauge outlet and by attaching either a thermocouple or thermometer to the cylinder wall. The gauge shall have scale divisions not greater than 10 psi for service pressures of less than or equal to 3000 psig. For service pressures greater than 3000 psig, a maximum of 100 psi per scale division is acceptable. If a thermometer is used, tape or putty shall be applied to the bulb to protect it from extraneous temperature influences, but do not apply putty between bulb and cylinder wall. The thermometer shall not have scale divisions greater than 2°F or 1°C. The containers shall be stabilized to ambient temperature. Then the valve shall be opened and the internal pressure observed on the gauge.

4.6.2 Leakage. Each deuterium container shall be tested for leaks at the neck threads, stem packing, and safety device of the valve with leak detection fluid. Valve seat leakage shall be tested after filling has been completed. This shall be done by applying the leak-detection fluid that leaves no residue on the outlet.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product furnished under this specification is a hazardous material as defined and regulated by the Department of Transportation (DOT) regulations. All packaging to be shipped commercially by any mode of transportation shall comply with the requirements of DOT regulations 49 CFR by the shipper in conjunction with the Commander, Headquarters Military Traffic Management and Terminal Service, ATTN: Safety Division/TES, Wash DC 20315. All packaging to be shipped by military air shall comply with AFR-71-4. The product shall be packaged in containers and unit quantities as specified by the procuring activity (6.2 and 6.3).

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5.2 Preparation of Containers. The contractor shall establish the condition of all containers and insure that 5.3 and 5.4, as applicable, are complied with. Contractor owned containers shall be cleaned, tested, and repaired in accordance with the schedule established in the contract or purchase order (6.2).

5.3 Cylinders. Cylinders shall be prepared in accordance with MIL-STD-1411.

5.4 Labeling and Marking. Each container shall be labeled and placarded in accordance with MIL-STD-129 and established DOT regulations or special permit. In addition, an identification tag and container color code shall be used.

5.4.1 Identification Tag. Unless otherwise specified in the contract or purchase order, an identification tag impervious to climatic conditions shall be wired to the outlet port of each container and shall contain the following information: Product name, specification number with revision letter, type designation, FSN number, quantity, name of manufacturer, name of contractor (if different from the manufacturer), date of manufacture, and lot identification number (6.2).

5.4.2 Cylinder Color Code. Unless otherwise specified by the procuring activity, each cylinder shall be color coded in accordance with MIL-STD-101 and the exact name identification to be marked on the outside of the cylinders shall be "Deuterium" (6.2). Any other name identification shall be obliterated by removing or overpainting.

5.5 Filter. A filter with a 10 micron nominal and 40 micron absolute rating shall be installed between the manufacturer's plant system and the manifold used to fill the gas containers for delivery.

6. NOTES

6.1 Intended Use. The deuterium propellant covered by this specification is intended for use as a fuel in Laser systems.

6.2 Ordering Data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following.

6.2.1 Procurement Requirements.

(a) Title, number, type, and date of this specification.

(b) Method of shipment, type and capacity of containers (5.1 and 6.3).

(c) The basis for procurement should be the cubic foot based on normal atmospheric conditions (14.7 psia and 70°F).

(d) When inspection requirements are to be performed by other than the supplier (4.1).

(e) When variation in points of inspection is granted (4.1.1).

(f) When waiver of quality conformance tests on each sample is granted (4.3.2.2).

(g) When disposition of rejected product has to be specified (4.4).

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(h) When cleaning and repair schedule is required for leased or Government-owned containers (5.2).

(i) When instructions for disposition of rejected cylinders are required.

(j) When identification tag is to be other than specified (5.4.1).

(k) When the container color code is to be waived (5.4.2).

6.3 Containers. As of the date of this specification, the following listed containers are considered acceptable for military use and are approved for use by DOT regulations as specified in 49 CFR 170-190.

(a) Sample quantities in cylinders per MIL-C-83690, Type II, Style 2, high pressure, 3000 ml.
(b) Cylinders of specifications DOT 3A, 3AA, 3AX, and 3AAX.

6.4 Safety. The following safety considerations should be applied as appropriate in all production testing, handling, and storage operations.

6.4.1 Flammability. Hydrogen (Deuterium) has an extremely wide flammability range (explosive limits). The lower and upper limits by volume are 4 and 75 percent. Working areas should be continuously monitored. Smoking or open flames in the vicinity where propellant is being handled or stored should not be permitted. Electrical grounds should be mandatory on all test equipment. Test personnel should wear nonstatic producing clothing.

6.4.2 Ventilation. Deuterium is highly flammable and can cause asphyxiation when the oxygen content of the air is reduced below 16 percent by volume. Where possible, all handling and storage of deuterium should be conducted in an open area. Ventilation should be mandatory in all closed areas where the propellant is handled.

6.4.3 Gas Cylinder Inspection. To prevent the possibility of electrical ignition of residual deuterium gas in the cylinder once the valve has been removed, no drop light inspection on the interior of the hydrogen cylinder should be made without first purging the interior of the cylinder with nitrogen gas for at least fifteen minutes with a flow rate of not less than two standard cubic feet per minute.

6.4.4 Highway Safety. To promote safety in the transportation of deuterium in interstate commerce by motor vehicle, the shipper should assure that each driver possesses a copy of Emergency Response Guide Book DOT P5800.2 guide-sheet on hydrogen.

6.5 Pollution Control. US Public Laws dictate increased effort to improve air, land, and water pollution control of toxic propellant vapors, leaks, spills, and disposal during all phases of manufacture, transfer, storage and transportation operations. The manufacturer/supplier is enjoined to approach the appropriate pollution control district to mutually resolve all problem areas and to develop adequate control and disposal methods for situations which are likely to develop in any of the phases.

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6.6 Supercession Data. This specification includes the requirements of the following Purchase Description

AFPID	9135-20	Dated	5 September 1972

Custodian:

Air Force - 12
Army - MI
Navy - AS

Preparing Activity:

Air Force - 12

Review Activities:

Air Force - 68

(Project 9135-F078)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-P-87847

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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